

NCERT Exemplar Solutions For Grade 12

Mathematics

Chapter 1: Relations and Functions

Solved Examples

Example 1 Let $A = \{0, 1, 2, 3\}$ and define a relation R on A as follows:

$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$. Is R reflexive? symmetric? Transitive?

Ans: R is reflexive and symmetric, but not transitive

Since, for $(1, 0) \in R$ and $(0, 3) \in R$ whereas $(1, 3) \notin R$.

Example 2 For the set $A = \{1, 2, 3\}$, define a relation R in the set A as follows: $R = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$. Write the ordered pairs to be added to R to make it is the smallest equivalence relation.

Ans: $(3, 1)$ is the single ordered pair which needs to be added to R to make it the smallest equivalence relation.

Example 3 Let R be the equivalence relation in the set Z of integers given by $R = \{(a, b): 2 \text{ divides } a - b\}$. Write the equivalence class $[0]$.

Ans: $[0] = \{0, \pm 2, \pm 4, \pm 6, \dots\}$.

Example 4 Let the function $f : R \rightarrow R$ be defined by $f(x) = 4x - 1, \forall x \in R$. Then, show that f is one-one.

Ans: For any two elements $x_1, x_2 \in R$ such that $f(x_1) = f(x_2)$,

we have $4x_1 - 1 = 4x_2 - 1$

$\Rightarrow 4x_1 = 4x_2$, i.e., $x_1 = x_2$

Hence, f is one-one.

Example 5 If $f = \{(5, 2), (6, 3)\}$, $g = \{(2, 5), (3, 6)\}$, write $f \circ g$.

Ans: Since, $f \circ g = f(g(x))$

Now, $f(g(2))=f(5)=2$ and $f(g(3))=f(6)=3$

Thus, $f \circ g = \{(2,2), (3,3)\}$.

Example 6 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = 4x - 3 \forall x \in \mathbb{R}$. Then write f^{-1} .

Ans: Given that $f(x) = 4x - 3 = y$ (say), then

$$4x = y + 3$$

$$x = \frac{y + 3}{4}$$

$$\text{Hence } f^{-1}(y) = \frac{y + 3}{4}$$

$$\Rightarrow f^{-1}(x) = \frac{x + 3}{4}$$

Example 7 Is the binary operation $*$ defined on $\mathbb{Z}$ (set of integer) by $m * n = m - n + mn \forall m, n \in \mathbb{Z}$ commutative?

Ans: No. Since for $1, 2 \in \mathbb{Z}$, $1 * 2 = 1 - 2 + 1.2 = 1$ while $2 * 1 = 2 - 1 + 2.1 = 3$ so that $1 * 2 \neq 2 * 1$.

Example 8 If $f = \{(5, 2), (6, 3)\}$ and $g = \{(2, 5), (3, 6)\}$, write the range of f and g .

Ans: The range of $f = \{2, 3\}$ and the range of $g = \{5, 6\}$.

Example 9 If $A = \{1, 2, 3\}$ and f, g are relations corresponding to the subset of $A \times A$ indicated against them, which of f, g is a function? Why?

$$f = \{(1, 3), (2, 3), (3, 2)\}$$

$$g = \{(1, 2), (1, 3), (3, 1)\}$$

Ans: f is a function since each element of A in the first place in the ordered pairs is related to only one element of A in the second place while g is not a function because 1 is related to more than one element of A , namely, 2 and 3.

Example 10 If $A = \{a, b, c, d\}$ and $f = \{(a, b), (b, d), (c, a), (d, c)\}$, show that f is one one from A onto A . Find f^{-1} .

Ans: f is one-one since each element of A is assigned to a distinct element of the set A . Also, f is onto since $f(A) = A$.

Moreover, $f^{-1} = \{(b, a), (d, b), (a, c), (c, d)\}$.

Example 11 In the set N of natural numbers, define the binary operation $*$ by $m * n = \text{g.c.d}(m, n)$, $m, n \in N$. Is the operation $*$ commutative and associative?

Ans: The operation is clearly commutative since

$$m * n = \text{g.c.d}(m, n) = \text{g.c.d}(n, m) = n * m \quad \forall m, n \in N.$$

It is also associative because for $l, m, n \in N$, we have

$$l * (m * n) = \text{g.c.d}(l, \text{g.c.d}(m, n))$$

$$= \text{g.c.d}(\text{g.c.d}(l, m), n)$$

$$= (l * m) * n.$$

Example 12 In the set of natural numbers N , define a relation R as follows: $\forall n, m \in N$, nRm if on division by 5 each of the integers n and m leaves the remainder less than 5, i.e. one of the numbers 0, 1, 2, 3 and 4. Show that R is an equivalence relation. Also, obtain the pairwise disjoint subsets determined by R .

Ans: R is reflexive since for each $a \in N$, aRa . R is symmetric since if aRb , then bRa for $a, b \in N$. Also, R is transitive since for $a, b, c \in N$, if aRb and bRc , then aRc . Hence R is an equivalence relation in N which will partition the set N into the pairwise disjoint subsets.

The equivalent classes are as mentioned below:

$$A_0 = \{5, 10, 15, 20 \dots\}$$

$$A_1 = \{1, 6, 11, 16, 21 \dots\}$$

$$A_2 = \{2, 7, 12, 17, 22, \dots\}$$

$$A_3 = \{3, 8, 13, 18, 23, \dots\}$$

$$A_4 = \{4, 9, 14, 19, 24, \dots\}$$

It is evident that the above five sets are pairwise disjoint and

$$A_0 \cup A_1 \cup A_2 \cup A_3 \cup A_4 = \bigcup_{i=0}^4 A_i = \mathbb{N}.$$

Example 13 Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2 + 1}$, $\forall x \in \mathbb{R}$, is neither one-one nor onto.

Ans: For $x_1, x_2 \in \mathbb{R}$, consider

$$f(x_1) = f(x_2)$$

$$\Rightarrow \frac{x_1}{x_1^2 + 1} = \frac{x_2}{x_2^2 + 1}$$

$$\Rightarrow x_1 x_2^2 + x_1 = x_2 x_1^2 + x_2$$

$$\Rightarrow x_1 x_2 (x_2 - x_1) = x_2 - x_1$$

$$\Rightarrow x_1 = x_2 \text{ or } x_1 x_2 = 1$$

We note that there are point, x_1 and x_2 with $x_1 \neq x_2$ and $f(x_1) = f(x_2)$, for instance, if we take $x_1 = 2$ and $x_2 = \frac{1}{2}$, then we have $f(x_1) = \frac{2}{5}$ and

$$f(x_2) = \frac{2}{5} \text{ but } 2 \neq \frac{1}{2}.$$

Hence f is not one-one. Also, f is not onto for if so then for $1 \in \mathbb{R} \exists x \in \mathbb{R}$ such that $f(x) = 1$

Which gives $\frac{x}{x^2 + 1} = 1$. But there is no such x in the domain $\mathbb{R}$, since the equation $x^2 - x + 1 = 0$ does not give any real value of x .

Example 14 Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined as $f(x) = |x| + x$ and $g(x) = |x| - x \forall x \in \mathbb{R}$. Then, find $f \circ g$ and $g \circ f$.

Ans: Here $f(x) = |x| + x$ which can be redefined as

$$f(x) = \begin{cases} 2x, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

$$0, \text{ if } x < 0$$

Similarly, the function g defined by $g(x) = |x| - x$ may be redefined as $g(x) = \begin{cases} 0 & \text{if } x \geq 0 \\ -2x & \text{if } x < 0 \end{cases}$

Therefore, $g \circ f$ gets defined as:

$$\text{For } x \geq 0, (g \circ f)(x) = g(f(x)) = g(2x) = 0$$

$$\text{and for } x < 0, (g \circ f)(x) = g(f(x)) = g(0) = 0.$$

Consequently, we have $(g \circ f)(x) = 0, \forall x \in \mathbb{R}$.

Similarly, $f \circ g$ gets defined as:

$$\text{For } x \geq 0, (f \circ g)(x) = f(g(x)) = f(0) = 0, \text{ and for } x < 0, (f \circ g)(x) = f(g(x)) = f(-2x) = -4x.$$

$$\text{i.e. } (f \circ g)(x) = \begin{cases} 0, & x \geq 0 \\ -4x, & x < 0 \end{cases}$$

Example 15 Let $\mathbb{R}$ be the set of real numbers and $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = 4x + 5$. Show that f is invertible and find f^{-1} .

Ans: Here the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = 4x + 5 = y$ (say).

$$\text{Then, } 4x = y - 5 \text{ or } x = \frac{y-5}{4}$$

This leads to a function $g: \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$g(y) = \frac{y-5}{4}$$

$$\text{Therefore, } (g \circ f)(x) = g(f(x)) = g(4x + 5) = \frac{4x + 5 - 5}{4} = x$$

$$\text{Or, } g \circ f = I_{\mathbb{R}}$$

$$\text{Similarly } (f \circ g)(y) = f(g(y)) = f\left(\frac{y-5}{4}\right) = 4\left(\frac{y-5}{4}\right) + 5 = y$$

$$\text{Or, } f \circ g = I_{\mathbb{R}}$$

Hence f is invertible and $f^{-1} = g$ which is given by

$$f^{-1}(x) = \frac{x-5}{4}.$$

Example 16 Let $*$ be a binary operation defined on Q . Find which of the following binary operations are associative

(i) $a * b = a - b$ for $a, b \in Q$.

Ans: (i) $*$ is not associative for if we take $a = 1$, $b = 2$ and $c = 3$, then $(a * b) * c = (1 * 2) * 3 = (1 - 2) * 3 = -1 - 3 = -4$ and $a * (b * c) = 1 * (2 * 3) = 1 * (2 - 3) = 1 - (-1) = 2$.

Thus $(a * b) * c \neq a * (b * c)$ and hence $*$ is not associative.

(ii) $a * b = \frac{ab}{4}$ for $a, b \in Q$.

Ans: (ii) $*$ is associative since Q is associative with respect to multiplication.

(iii) $a * b = a - b + ab$ for $a, b \in Q$.

Ans: (iii) $*$ is not associative for if we take $a = 2$, $b = 3$ and $c = 4$, then

$$(a * b) * c = (2 * 3) * 4 = (2 - 3 + 6) * 4 = 5 * 4 = 5 - 4 + 20 = 21, \text{ and}$$

$$a * (b * c) = 2 * (3 * 4) = 2 * (3 - 4 + 12) = 2 * 11 = 2 - 11 + 22 = 13$$

Thus, $(a * b) * c \neq a * (b * c)$ and hence $*$ is not associative.

(iv) $a * b = ab^2$ for $a, b \in Q$.

Ans: (iv) $*$ is not associative for if we take $a = 1$, $b = 2$ and $c = 3$, then

$$(a * b) * c = (1 * 2) * 3 = 4 * 3 = 4 \times 9 = 36 \text{ and}$$

$$a * (b * c) = 1 * (2 * 3) = 1 * 18 = 1 \times 182 = 324.$$

Thus $(a * b) * c \neq a * (b * c)$ and hence $*$ is not associative.

Objective Type Questions

Choose the correct answer from the given four options in each of the Examples 17 to 25.

Example 17 Let R be a relation on the set N of natural numbers defined by nRm if n divides m . Then R is

- (A) Reflexive and symmetric
- (B) Transitive and symmetric
- (C) Equivalence
- (D) Reflexive, transitive but not symmetric

Ans: The correct choice is (D).

Since n divides n , $\forall n \in N$, R is reflexive.

R is not symmetric since for $3, 6 \in N$, $3R6 \neq 6R3$.

R is transitive since for n, m, r whenever n/m and $m/r \Rightarrow n/r$, i.e., n divides m and m divides r , then n will divide r .

Example 18 Let L denote the set of all straight lines in a plane. Let a relation R be defined by lRm if and only if l is perpendicular to $m \forall l, m \in L$. Then R is

- (A) reflexive
- (B) symmetric
- (C) transitive
- (D) none of these

Ans: The correct choice is (B).

Example 19 Let N be the set of natural numbers and the function $f : N \rightarrow N$ be defined by $f(n) = 2n + 3 \forall n \in N$. Then f is

- (A) surjective
- (B) injective
- (C) bijective
- (D) none of these

Ans: (B) is the correct option

Example 20 Set A has 3 elements and set B has 4 elements. Then the number of injective mappings that can be defined from A to B is

- (A) 144
- (B) 12
- (C) 24
- (D) 64

Ans: The correct choice is (C). The total number of injective mappings from the set containing 3 elements into the set containing 4 elements is ${}^4P_3 = 4! = 24$

Example 21 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sin x$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = x^2$, then $f \circ g$ is

- (A) $x^2 \sin x$
- (B) $(\sin x)^2$
- (C) $\sin x^2$
- (D) $\frac{\sin x}{x^2}$

Ans: (C) is the correct choice.

Example 22 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x - 4$. Then $f^{-1}(x)$ is given by

- (A) $\frac{x+4}{3}$
- (B) $\frac{x}{3} - 4$
- (C) $3x + 4$
- (D) None of these

Ans: (C) is the correct choice.

Example 23 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 + 1$. Then, pre-images of 17 and -3, respectively, are

- (A) $\varnothing, \{4, -4\}$
- (B) $\{3, -3\}, \varnothing$
- (C) $\{4, -4\}, \varnothing$
- (D) $\{4, -4, \{2, -2\}\}$

Ans: (C) is the correct choice since for $f^{-1}(17) = x$

$$\Rightarrow f(x) = 17 \text{ or } x^2 + 1 = 17$$

$$\Rightarrow x = \pm 4 \text{ or } f^{-1}(17) = \{4, -4\} \text{ and for } f^{-1}(-3) = x$$

$$\Rightarrow f(x) = -3 \Rightarrow x^2 + 1 = -3$$

$$\Rightarrow x^2 = -4 \text{ and hence } f^{-1}(-3) = \varnothing.$$

Example 24 For real numbers x and y , define xRy if and only if $x - y + \sqrt{2}$ is an irrational number. Then the relation R is

- (A) reflexive
- (B) symmetric
- (C) transitive
- (D) none of these

Ans: (A) is the correct choice.

Example 25 Consider the set $A = \{1, 2, 3\}$ and R be the smallest equivalence relation on A , then $R =$ _____

Ans: $R = \{(1, 1), (2, 2), (3, 3)\}.$

Example 26 The domain of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sqrt{x^2 - 3x + 2}$ is _____.

Ans: Here $x^2 - 3x + 2 \geq 0$

$$\Rightarrow (x-1)(x-2) \geq 0$$

$$\Rightarrow x \leq 1 \text{ or } x \geq 2$$

Hence, the domain of $f = (-\infty, 1] \cup [2, \infty).$

Example 27 Consider the set A containing n elements. Then, the total number of injective functions from A onto itself is _____.

Ans: $n!$

Example 28 Let Z be the set of integers and R be the relation defined in Z such that aRb if $a - b$ is divisible by 3. Then R partitions the set Z into _____ pairwise disjoint subsets.

Ans: Three.

Example 29 Let R be the set of real numbers and $*$ be the binary operation defined on R as $a * b = a + b - ab \forall a, b \in R$. Then, the identity element with respect to the binary operation $*$ is _____.

Ans: 0 is the identity element with respect to the binary operation $*$.

State True or False for the statements in each of the Examples 30 to 34

Example 30 Consider the set $A = \{1, 2, 3\}$ and the relation

$R = \{(1, 2), (1, 3)\}$. R is a transitive relation.

Ans: True.

Example 31 Let A be a finite set. Then, each injective function from A into itself is not surjective.

Ans: False.

Example 32 For sets A , B and C , let $f : A \rightarrow B$, $g : B \rightarrow C$ be functions such that $g \circ f$ is injective. Then both f and g are injective functions.

Ans: False.

Example 33 For sets A , B and C , let $f : A \rightarrow B$, $g : B \rightarrow C$ be functions such that $g \circ f$ is surjective. Then g is surjective

Ans: True.

Example 34 Let N be the set of natural numbers. Then, the binary operation $*$ in N defined as $a * b = a + b, \forall a, b \in N$ has identity element.

Ans: False.

Short Answer Questions

1. Let $A = \{a, b, c\}$ and the relation R be defined on A as follows:

$$R = \{(a, a), (b, c), (a, b)\}.$$

Then, write the minimum number of ordered pairs to be added in R to make R reflexive and transitive.

Ans: Give relation, $R = \{(a, a), (b, c), (a, b)\}$

To make R reflexive we must add (b, b) and (c, c) to R .

Also, to make R transitive we must add (a, c) to R .

So, the minimum number of ordered pairs to be added is 3.

2. Let D be the domain of the real valued function f defined by $f(x) = \sqrt{25 - x^2}$. Then, write D .

Ans: Given real valued function $f(x)$, such that $f(x) = \sqrt{25 - x^2}$, since, $f(x)$ is real valued.

We must have

$$25 - x^2 \geq 0$$

$$\Rightarrow x^2 \leq 25$$

$$\Rightarrow -5 \leq x \leq 5$$

The domain is $D = [-5, 5]$

3. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x + 1$ and $g(x) = x^2 - 2, \forall x \in \mathbb{R}$, respectively. Then, find $g \circ f$.

Ans: Given that $f(x) = 2x + 1$ and $g(x) = x^2 - 2, \forall x \in \mathbb{R}$.

Then $(g \circ f)x = g\{f(x)\} = g(2x + 1) = (2x + 1)^2 - 2$

$$= 4x^2 + 4x + 1 - 2$$

$$= 4x^2 + 4x - 1$$

4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = 2x - 3 \forall x \in \mathbb{R}$. write f^{-1} .

Ans: Given $f(x) = 2x - 3 \forall x \in \mathbb{R}$

Now, let $a, b \in \mathbb{R}$ such that

$$f(a) = f(b)$$

$$\Rightarrow 2a - 3 = 2b - 3$$

$$\Rightarrow 2a = 2b$$

$\Rightarrow f(x)$ is one- one.

Also, if $x, y \in \mathbb{R}$, such that,

$$f(x) = y \Rightarrow 2x - 3 = y$$

$$\Rightarrow x = \frac{y + 3}{2} = g(y) \forall y \in \mathbb{R}$$

$\Rightarrow f(x)$ is onto and therefore is bijective implies $f(x)$ has an inverse

$$f^{-1}(x) = g(x) = \frac{x + 3}{2} \forall x \in \mathbb{R}$$

5. If $A = \{a, b, c, d\}$ and the function $f = \{(a, b), (b, d), (c, a), (d, c)\}$, write f^{-1} .

Ans: Let $f: A \rightarrow A$, then inverse f^{-1} is such that

$$f^{-1} = \{(b, a), (d, b), (a, c), (c, d)\}$$

$$f^{-1}: A \rightarrow A \text{ and } f^{-1} = \{(b, a), (d, b), (a, c), (c, d)\}$$

6. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^2 - 3x + 2$, write $f(f(x))$.

Ans: Given that $f(x) = x^2 - 3x + 2$

$$f(f(x)) = f(x^2 - 3x + 2)$$

$$= (x^2 - 3x + 2)^2 - 3(x^2 - 3x + 2) + 2$$

$$= x^4 + 9x^2 + 4 - 6x^3 - 12x + 4x^2 - 3x^2 + 9x - 6 + 2$$

$$= x^4 + 10x^2 - 6x^3 - 3x$$

$$f(f(x)) = x^4 + 10x^2 - 6x^3 - 3x$$

7. Is $g = \{(1,1), (2,3), (3,5), (4,7)\}$ a function? If g is described by $g(x) = \alpha x + \beta$, then what value should be assigned to α and β .

Ans: Yes g is a function since every element in the domain has a unique image.

$$\text{Now, let } g(x) = \alpha x + \beta$$

Then given,

$$g(1) = \alpha + \beta = 1$$

$$g(2) = 2\alpha + \beta = 3$$

Subtracting $g(1)$ from $g(2)$ gives,

$$(2\alpha + \beta) - (\alpha + \beta) = \alpha = 2 \text{ and substituting it into } g(1)$$

$$\text{We have } \beta = -1.$$

So, values of α and β are 2 and -1.

8. Are the following set of ordered pair's functions? If so, examine whether the mapping is injective or subjective.

(i) $\{(x,y) : x \text{ is a person, } y \text{ is the mother of } x\}$

Ans: Given set of ordered pairs is:

$$\{(x,y) : x \text{ is a person, } y \text{ is the mother of } x\}.$$

It represents a function. Here the image of distinct elements of x under f are not distinct, so it's not injective but it is subjective.

(ii) $\{(a,b) : a \text{ is a person , } b \text{ is the ancestor of } a\}$

Ans: Given set of ordered pairs is:

$\{(a, b) : a \text{ is a person, } b \text{ is the ancestor of } a\}$

Here, each element of the domain does not have a unique image.

So, it does not represent function.

9.If the mapping f and g are given by $f = \{(1,2),(3,5), (4,1)\}$ and $g = \{(2,3), (5,1), (1,3)\}$, write fog.

Ans: Given that,

$$f = \{(1,2),(3,5), (4,1)\}$$

$$\text{And } g = \{(2,3), (5,1), (1,3)\}$$

$$\text{Now, } fog(2) = f\{g(2)\} = f(3) = 5$$

$$fog(5) = f\{g(5)\} = f(1) = 2$$

$$fog(1) = f\{g(1)\} = f(3) = 5$$

$$fog = \{(2,5),(5,2),(1,5)\}.$$

10. Let C be the set of complex numbers. Prove that the mapping $f: C \rightarrow R$ given by $f(z) = |z|, \forall z \in C$, is neither one-one nor onto.

Ans: The mapping $f: C \rightarrow R$

$$\text{Given, } f(z) = |z|, \forall z \in C$$

$$f(1) = |1| = 1$$

$$f(-1) = |-1| = 1$$

$$f(1) = f(-1)$$

$$\text{But, } 1 \neq -1$$

So, $f(z)$ is not one-one. Also, $f(z)$ is not onto as there is no pre-image for any negative element of R under the mapping $f(z)$.

11. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \cos x$, $\forall x \in \mathbb{R}$. Show that f is neither one-one nor onto.

Ans: Given function $f(x) = \cos x$, $\forall x \in \mathbb{R}$

$$\text{Now, } f\left(\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0$$

$$f\left(\frac{-\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0$$

$$f\left(\frac{\pi}{2}\right) = f\left(\frac{-\pi}{2}\right)$$

$$f\left(\frac{-\pi}{2}\right) = f\left(\frac{\pi}{2}\right)$$

$$\text{But, } \frac{\pi}{2} \neq \frac{-\pi}{2} = 0$$

So, $f(x)$ is not one-one.

Now, $f(x) = \cos x$, $\forall x \in \mathbb{R}$ is not onto as there is no pre-image for any real number, which does not belong to the intervals $[-1, 1]$, the range of $\cos x$.

12. Let $X = \{1, 2, 3\}$ and $Y = \{4, 5\}$. Find whether the following subsets of $X \times Y$ are functions from X to Y or not.

(i) $f = \{(1, 4), (1, 5), (2, 4), (3, 5)\}$

Ans: f is not a function because f has no unique image.

(ii) $g = \{(1, 4), (2, 4), (3, 4)\}$

Ans: g is a function as each element of the domain has a unique image.

(iii) $h = \{(1, 4), (2, 5), (3, 5)\}$

Ans: h is a function

(iv) $k = \{(1, 4), (2, 5)\}$.

Ans: k is not a function as 3 does not have any image under the mapping.

13. If functions $f: A \rightarrow B$ and $g: B \rightarrow A$ satisfy $g \circ f = I_A$, then show that f is one-one and g is onto.

Ans: Given that $f: A \rightarrow B$ and $g: B \rightarrow A$ satisfy $g \circ f = I_A$

Since, $g \circ f = I_A$,

$$\Rightarrow g \circ \{f(x_1)\} = g \circ \{f(x_2)\}$$

$$\Rightarrow g(x_1) = g(x_2) \text{ [Since } g \circ f = I_A \text{]}$$

$$\Rightarrow x_1 = x_2$$

Hence, f is one-one and g is onto.

14. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined $f(x) = \frac{1}{2 - \cos x}$, $\forall x \in \mathbb{R}$. Then find the range of f .

Ans: Given function, $f(x) = \frac{1}{2 - \cos x}$, $\forall x \in \mathbb{R}$

$$\text{Let } y = \frac{1}{2 - \cos x}$$

$$\Rightarrow 2y - y \cos x = 1$$

$$\Rightarrow y \cos x = 2y - 1$$

$$\Rightarrow \cos x = \frac{2y - 1}{y} = 2 - \frac{1}{y}$$

$$\Rightarrow \cos x = 2 - \frac{1}{y}$$

$$\Rightarrow -1 \leq \cos x \leq 1$$

$$\Rightarrow -1 \leq 2 - \frac{1}{y} \leq 1$$

$$\Rightarrow -3 \leq -\frac{1}{y} \leq -1$$

$$\Rightarrow 3 \geq \frac{1}{y} \geq 1$$

$$\Rightarrow \frac{1}{3} \leq y \leq 1$$

So, the range of y , that is $f(x)$ is $\left[\frac{1}{3}, 1\right]$.

15. Let n be a fixed positive integer. Define a relation R in Z as follows: $\forall a, b \in Z$, aRb if and only if $a - b$ is divisible by n . Show that R is an equivalence relation.

Ans: Given that,

$\forall a, b \in Z$, aRb if and only if $a - b$ is divisible by n .

Now,

Reflexive

$aRa \Rightarrow (a-a)$ is divisible by n , which is true for any integer ' a ' as ' 0 ' is divisible by n .
Hence, r is reflexive.

Symmetric

aRb

$\Rightarrow a - b$ is divisible by n .

$\Rightarrow -b + a$ is divisible by n .

$\Rightarrow -(b - a)$ is divisible by n .

$\Rightarrow (b - a)$ is divisible by n .

bRa

Hence, R is symmetric.

Transitive

Let aRb and bRc

$\Rightarrow (a - b)$ is divisible by n and $(b - c)$ is divisible by n .

$\Rightarrow (a - b) + (b - c)$ is divisible by n .

$\Rightarrow (a - c)$ is divisible by n .

aRc

Hence, R is transitive.

So, R is an equivalence relation.

Long Answers:

16. If $A = \{1, 2, 3, 4\}$, define relations on A which have properties of being:

(a) reflexive, transitive but not symmetric

Ans: Given that: $A = \{1, 2, 3, 4\}$

Let $R_1 = \{(1, 1), (1, 2), (2, 3), (2, 2), (1, 3), (3, 3)\}$

R_1 is reflexive, since $(1, 1), (2, 2), (3, 3)$ lies in R_1

Now, $(1, 2) \in R_1, (2, 3) \in R_1 \Rightarrow (1, 3) \in R_1$

Hence, R_1 is also transitive but $(1, 2) \in R_1 \Rightarrow (2, 1) \notin R_1$

So, it is not symmetric.

(b) symmetric but neither reflexive nor transitive

Ans: Given that: $A = \{1, 2, 3, 4\}$

Let $R_2 = \{(1, 2), (2, 1)\}$

Now, $(1, 2) \in R_2, (2, 1) \in R_2$

So, it is symmetric.

(c) reflexive, symmetric and transitive.

Ans: Given that: $A = \{1, 2, 3, 4\}$

Let $R_3 = \{(1, 2), (2, 1), (1, 1), (2, 2), (3, 3), (1, 3), (3, 1), (2, 3)\}$

Hence, R_3 is reflexive, symmetric and transitive.

17. Let R be relation defined on the set of natural numbers N as follows:

$R = \{(x, y): x \in N, y \in N, 2x + y = 41\}$.

Find the domain and range of the relation R . Also verify whether R is reflexive, symmetric and transitive.

Ans: Given that: $R = \{(x, y): x \in N, y \in N, 2x + y = 41\}$.

Domain = $\{1, 2, 3, \dots, 20\}$

Range = $\{(1, 39), (2, 37), (3, 35), \dots, (19, 3), (20, 1)\}$

R is not reflexive as $(2, 2) \notin R$

$2 \times 2 + 2 \neq 41$

So, R is not symmetric.

As, $(1, 39) \in R$ but $(39, 1) \notin R$

So, R is not transitive

As $(11, 19) \in R$, $(19, 3) \in R$ but $(11, 3) \notin R$

Hence, R is neither reflexive, nor symmetric nor transitive.

18. Given $A = \{2, 3, 4\}$, $B = \{2, 5, 6, 7\}$. Construct an example of each of the following:

(a) an injective mapping from A to B

Ans: Given that, $A = \{2, 3, 4\}$, $B = \{2, 5, 6, 7\}$.

Let $f : A \rightarrow B$ denote a mapping

$f = \{(x, y) : y = x + 3\}$

I.e $f = \{(2, 5), (3, 6), (4, 7)\}$

Which is an injective mapping.

(b) a mapping from A to B which is not injective

Ans: Given that $A = \{2, 3, 4\}$, $B = \{2, 5, 6, 7\}$.

Let $g : A \rightarrow B$ denote a mapping

Such that $g = \{(2, 2), (3, 5), (4, 7)\}$ which is not an injective mapping.

(c) a mapping from B to A.

Ans: Given that $A = \{2, 3, 4\}$, $B = \{2, 5, 6, 7\}$.

Let $h : B \rightarrow A$ denote a mapping

Such that $h = \{(2, 2), (5, 3), (6, 4), (7, 4)\}$ which is a mapping from $B \rightarrow A$.

19. Give an example of a map

(i) which is one-one but not onto

Ans: Let $f: \mathbb{N} \rightarrow \mathbb{N}$, be a mapping defined by $f(x) = 2x$ which is one-one

For $f(x_1) = f(x_2)$

$$\Rightarrow 2x_1 = 2x_2$$

$$\Rightarrow x_1 = x_2$$

Further f is not onto, as for $1 \in \mathbb{N}$, there does not exist any x in $\mathbb{N}$ such that $f(x) = 2x + 1$.

(ii) which is not one-one but onto

Ans: Let $f: \mathbb{N} \rightarrow \mathbb{N}$, given by $f(1) = f(2) = 1$ and $f(x) = x - 1$ for every $x > 2$ is onto but not one-one. f is not one-one as $f(1) = f(2) = 1$. But f is onto.

(iii) which is neither one-one nor onto.

Ans: The mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$, is neither one -one nor onto.

20. Let $A = \mathbb{R} - \{3\}$, $B = \mathbb{R} - \{1\}$. Let $f: A \rightarrow B$ be defined by $f(x) = \frac{x-2}{x-3} \forall x \in A$.

Then show that f is bijective.

Ans: Given that, $A = \mathbb{R} - \{3\}$, $B = \mathbb{R} - \{1\}$.

$f: A \rightarrow B$ is defined by $f(x) = \frac{x-2}{x-3} \forall x \in A$.

For injectivity

Let $f(x_1) = f(x_2)$

$$\Rightarrow \frac{x_1 - 2}{x_1 - 3} = \frac{x_2 - 2}{x_2 - 3}$$

$$\Rightarrow (x_1 - 2)(x_2 - 3) = (x_2 - 2)(x_1 - 3)$$

$$\Rightarrow x_1 x_2 - 3 x_1 - 2 x_2 + 6 = x_1 x_2 - 3 x_2 - 2 x_1 + 6$$

$$\Rightarrow -3 x_1 - 2 x_2 = -3 x_2 - 2 x_1$$

$$\Rightarrow -x_1 = -x_2$$

$$\Rightarrow x_1 = x_2$$

So, $f(x)$ is an injective function.

For surjectivity,

$$\text{Let } y = \frac{x-2}{x-3}$$

$$\Rightarrow x - 2 = xy - 3y$$

$$\Rightarrow x(1-y) = 2 - 3y$$

$$\Rightarrow x = \frac{2-3y}{1-y}$$

$$\Rightarrow x = \frac{2-3y}{1-y} \in A \quad \forall y \in B \text{ (codomain)}$$

So, $f(x)$ is a surjective function.

Hence, $f(x)$ is a bijective function.

21. Let $A = [-1, 1]$. Then, discuss whether the following functions defined on A are one-one, onto or bijective:

(i) $f(x) = \frac{x}{2}$

Ans: Given : $A = [-1, 1]$

$$f(x) = \frac{x}{2}$$

$$\text{Let } f(x_1) = f(x_2)$$

$$\Rightarrow \frac{x_1}{2} = \frac{x_2}{2}$$

$$\Rightarrow x_1 = x_2$$

So, $f(x)$ is one-one.

$$\Rightarrow x = 2y \notin A, \forall y \in A$$

As for $y = 1 \in A$, $x = 2 \notin A$

So, $f(x)$ is not onto.

Also, $f(x)$ is not bijective as it is not onto.

(ii) $g(x) = |x|$

Ans: Let $g(x_1) = g(x_2)$

$$\Rightarrow |x_1| = |x_2|$$

$$\Rightarrow x_1 = \pm x_2$$

So, $g(x)$ is not one-one.

$$\text{Now, } y = |x| \Rightarrow x = \pm y \notin A, \forall y \in A$$

So, $g(x)$ is not onto also, $g(x)$ is not bijective.

(iii) $h(x) = x|x|$

Ans: $h(x) = x|x|$

$$\Rightarrow x_1|x_1| = x_2|x_2|$$

$$\Rightarrow x_1 = x_2$$

So, $h(x)$ is one-one.

$$\text{Now, let } y = x|x|$$

$$\Rightarrow y = x^2 \in A, \forall x \in A$$

So, $h(x)$ is onto also, $h(x)$ is bijective.

(iv) $k(x) = x^2$

$$\text{Let } k(x_1) = k(x_2)$$

$$\Rightarrow x_1^2 = x_2^2$$

$$\Rightarrow x_1 = \pm x_2$$

Thus, $k(x)$ is not one-one.

$$\text{Now, let } y = x^2$$

$$\Rightarrow x = \sqrt{y} \in A, \forall y \in A$$

$$\Rightarrow x = \sqrt{y} \notin A, \forall y \in A$$

$$\text{As for } y = -1, x = \sqrt{-1} \notin A$$

Hence, $k(x)$ is neither one-one nor onto.

22. Each of the following defines a relation on $\mathbb{N}$. Determine which of the above relations are reflexive, symmetric and transitive.

(i) x is greater than y , $x, y \in \mathbb{N}$

Ans: x is greater than y , $x, y \in \mathbb{N}$

$$(x, x) \in R$$

For xRx , $x > x$ is not true for any $x \in \mathbb{N}$

Therefore, R is not reflexive.

$$\text{Let } (x, y) \in R \Rightarrow xRy$$

$$x > y$$

But $y > x$ is not true for any $x, y \in \mathbb{N}$

Thus, R is not symmetric.

$$\text{Let } xRy \text{ and } yRz$$

$$x > y \text{ and } y > z \Rightarrow x > z$$

$$\Rightarrow xRz$$

So, r is transitive.

(ii) $x + y = 10$, $x, y \in \mathbb{N}$

Ans: $x + y = 10$, $x, y \in \mathbb{N}$

$$R = \{(x, y); x + y = 10, x, y \in \mathbb{N}\}$$

$$R = \{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1)\} (1, 1) \notin R$$

So, R is not reflexive.

$$(x, y) \in R \Rightarrow (y, x) \in R$$

Therefore, R is symmetric.

$$(1, 9) \in R, (9, 1) \in R \Rightarrow (1, 1) \notin R$$

Hence, R is not transitive.

(iii) xy is square of an integer $x, y \in \mathbb{N}$

Ans: Given xy is the square of the integer. $x, y \in \mathbb{N}$

$\Rightarrow R = \{(x,y): xy \text{ is a square of an integer } x, y \in \mathbb{N}\}$

$(x,x) \in R, \forall x \in \mathbb{N}$

As x^2 is square of an integer for any $x \in \mathbb{N}$

Hence, R is reflexive.

If $(x,y) \in R \Rightarrow (y,x) \in R$

Therefore, R is symmetric.

If $(x,y) \in R, (y,z) \in R$

So, xy is square of an integer and yz is square of an integer.

Let $xy = m^2$ and $yz = n^2$ for some $m, n \in \mathbb{Z}$

$$x = \frac{m^2}{y} \text{ and } z = \frac{n^2}{y}$$

$$xz = \frac{m^2 n^2}{y}, \text{ which is square of an integer.}$$

So, R is transitive.

(iv) $x + 4y = 10, x, y \in \mathbb{N}$.

Ans: $R = \{(x, y): x + 4y = 10, x, y \in \mathbb{N}\}$

$R = \{(2,2), (6,1)\}$

$(1,1), (3,3) \dots \notin R$

Thus, R is not reflexive.

$(6,1) \in R$ but $(1,6) \notin R$

Hence, R is not symmetric.

$(x,y) \in R \Rightarrow x + 4y = 10$ but $(y,z) \in R$

$y + 4z = 10 \Rightarrow (x, z) \in R$

So, R is transitive.

23. Let $A = \{1, 2, 3, \dots, 9\}$ and R be the relation in $A \times A$ defined by $(a, b) R (c, d)$ if $a + d = b + c$ for $(a, b), (c, d)$ in $A \times A$. Prove that R is an equivalence relation and also obtain the equivalent class $[(2, 5)]$.

Ans: Given that $A = \{1, 2, 3, \dots, 9\}$

$(a, b) R (c, d)$ if $a + d = b + c$ for $(a, b), (c, d) \in A \times A$

So, $(a, b) \in A \times A$

$(c, d) \in A \times A$

Let $(a, b) R (a, b)$

$\Rightarrow a + b = b + a, \forall a, b \in A$ which is true for any $a, b \in A$.

Hence, R is reflexive.

Let $(a, b) R (c, d)$

$a + b = b + c$

$c + b = d + a$

$\Rightarrow (c, d) R (a, b)$

So, R is symmetric.

Let $(a, b) R (c, d)$ and $(c, d) R (e, f)$

$\Rightarrow a + d = b + c$ and $c + f = d + e$

$\Rightarrow a + d = b + c$ and $d + e = c + f$

$\Rightarrow (a + d) - (d + e) = (b + c) - (c + f)$

$\Rightarrow (a - e) = b - f$

$\Rightarrow a + f = b + e$

$\Rightarrow (a, b) R (e, f)$

Now, equivalence class containing

$[(2, 5)]$ is $\{(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)\}$.

24. Using the definition, prove that the function $f : A \rightarrow B$ is invertible if and only if f is both one-one and onto.

Ans: A function $f: x \rightarrow y$ is defined to be invertible, if there exist a function

$g = y \rightarrow x$ such that $g \circ f = I_x$ and $f \circ g = I_y$. The function is called the inverse of f and is denoted by f^{-1} .

A function $f = x \rightarrow y$ is invertible if f is a bijective function.

25. Functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ are defined, respectively, by $f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$, find

(i) $f \circ g$

Ans: Given that $f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$

$$f \circ g = f\{g(x)\} = f(2x - 3)$$

$$= (2x - 3)^2 + 3(2x - 3) + 1$$

$$= 4x^2 + 9 - 12x + 6x - 9x + 1$$

$$= 4x^2 - 6x + 1$$

(ii) $g \circ f$

Ans: Given that $f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$

$$g \circ f = g\{f(x)\} = g(x^2 + 3x + 1)$$

$$= 2(x^2 + 3x + 1) - 3$$

$$= 2x^2 + 6x + 2 - 3$$

$$= 2x^2 + 6x - 1$$

(iii) $f \circ f$

Ans: Given that $f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$

$$f \circ f = f\{f(x)\} = f(x^2 + 3x + 1)$$

$$= (x^2 + 3x + 1)^2 + 3(x^2 + 3x + 1) + 1$$

$$= x^4 + 9x^2 + 1 + 6x^3 + 6x + 2x^2 + 3x^2 + 9x + 3 + 1$$

$$= x^4 + 6x^3 + 14x^2 + 15x + 5$$

(iv) $g \circ g$

Ans: Given that $f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$

$$g \circ g = g\{g(x)\} = g(2x - 3)$$

$$= 2(2x - 3) - 3$$

$$= 4x - 6 - 3$$

$$= 4x - 9$$

26. Let $*$ be the binary operation defined on Q . Find which of the following binary operations are commutative

(i) $a * b = a - b \forall a, b \in Q$

Ans: $a * b = a - b \forall a, b \in Q$ and $b * a = b - a$

So, $a * b \neq b * a$ [Since, $b - a \neq a - b$]

Hence, $*$ is not commutative.

(ii) $a * b = a^2 + b^2 \forall a, b \in Q$

Ans: $a * b = a^2 + b^2 \forall a, b \in Q$

$b * a = b^2 + a^2$

Hence, $*$ is commutative. [Since, $+$ is on rational is commutative].

(iii) $a * b = a + ab \forall a, b \in Q$

Ans: $a * b = a + ab \forall a, b \in Q$

$b * a = b + ab$

Clearly, $a + ab \neq b + ab$

Hence, $*$ is not commutative.

(iv) $a * b = (a - b)^2 \forall a, b \in Q$

Ans: $a * b = (a - b)^2 \forall a, b \in Q$

$b * a = (b - a)^2$

Since, $(a - b)^2 = (b - a)^2$

Hence, $*$ is commutative.

27. Let $*$ be binary operation defined on R by $a * b = 1 + ab, \forall a, b \in R$. Then the operation $*$ is

(i) commutative but not associative

(ii) associative but not commutative

(iii) neither commutative nor associative

(iv) both commutative and associative

Ans: Given that $a * b = 1 + ab, \forall a, b \in \mathbb{R}$

$$a * b = ab + 1 = b * a$$

So, $*$ is a commutative binary operation

$$\text{Also, } a * (b * c) = a * (1 + bc) = 1 + a(1 + bc)$$

$$a * (b * c) = 1 + a + abc \quad \dots\dots\dots (1)$$

$$(a * b) * c = (1 + ab) * c$$

$$= 1 + (1 + ab)c = 1 + c + abc \quad \dots\dots\dots (2)$$

From (1) and (2)

$$a * (b * c) \neq (a * b) * c$$

So, $*$ is not associative

Hence, $*$ is commutative but not associative.

Objective Type Questions

28. Let T be the set of all triangles in the Euclidean plane, and let a relation R on T be defined as aRb if a is congruent to $b \forall a, b \in T$. Then R is

- (A) reflexive but not transitive**
- (B) transitive but not symmetric**
- (C) equivalence**
- (D) none of these**

Ans:

Consider that aRb , if a is congruent to $b, \forall a, b \in T$.

Then, $aRa \Rightarrow a \sqcup a$,

Which is true for all $a \in T$.

So, R is reflexive----- (i)

Let $aRb \Rightarrow a \sqcup b$,

$\Rightarrow b \sqcup a$

$\Rightarrow bRa$

So, R is symmetric----- (ii)

Let aRb and bRc

$\Rightarrow a \sqcup b$ and $b \sqcup c$

$\Rightarrow a \sqcup c$

$\Rightarrow aRc$

So, R is transitive----- (iii)

Hence, R is an equivalence relation.

29. Consider the non-empty set consisting of children in a family and a relation R defined as aRb if a is brother of b. Then R is

- (A) symmetric but not transitive
- (B) transitive but not symmetric
- (C) neither symmetric nor transitive
- (D) both symmetric and transitive

Ans:

Given: $aRb \Rightarrow a$ is brother of b .

Therefore $aRa \Rightarrow a$ is brother of a , which is not true.

So, R is not reflexive.

$aRb \Rightarrow a$ is brother of b .

This doesn't mean b is also a brother of a as b can be sister of a .

Hence, R is not symmetric.

$aRb \Rightarrow a$ is brother of b .

And, $bRc \Rightarrow b$ is brother of c .

So, a is brother of c

Hence, R is transitive.

30. The maximum number of equivalence relations on the set $A = \{1, 2, 3\}$ are

(A) 1

(B) 2

(C) 3

(D) 5

Ans: Given that, $A = \{1, 2, 3\}$

Now, number of equivalence relations are as follows:

$$R_1 = \{(1, 1), (2, 2), (3, 3)\}$$

$$R_2 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (2, 2), (3, 3), (1, 3), (3, 1)\}$$

$$R_4 = \{(1, 1), (2, 2), (3, 3), (2, 3), (3, 2)\}$$

$$R_5 = \{(1, 2, 3) \Leftrightarrow A \times A = A^2\}$$

Maximum number of equivalence relation on the set $A = \{1, 2, 3\} = 5$.

31. If a relation R on the set $\{1, 2, 3\}$ be defined by $R = \{(1, 2)\}$, then R is

(A) reflexive

(B) transitive

(C) symmetric

(D) none of these

Ans: (B)

R on the set $\{1, 2, 3\}$ be defined by $R = \{(1, 2)\}$

It is clear that R is transitive.

32. Let us define a relation R in R as aRb if $a \geq b$. Then R is

(A) an equivalence relation

(B) reflexive, transitive but not symmetric

(C) symmetric, transitive but

(D) neither transitive nor reflexive, not reflexive but symmetric.

Ans: (B)

Given that, aRb if $a \geq b$

$\Rightarrow aRa \Rightarrow a \geq a$, which is true.

Let aRb , $a \geq b$, then $b \geq a$ which is not true R is not symmetric.

But aRb and bRc

$\Rightarrow a \geq b$ and $b \geq c$

$\Rightarrow a \geq c$

Hence, R is transitive.

33. Let $A = \{1, 2, 3\}$ and consider the relation

$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$. Then R is

(A) reflexive but not symmetric

(B) reflexive but not transitive

(C) symmetric and transitive

(D) neither symmetric, nor transitive.

Ans: Given that $A = \{1, 2, 3\}$

And $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$.

Since, $(1, 1), (2, 2), (3, 3) \in R$

Hence, R is reflexive.

$(1, 2) \in R$ but $(2, 1) \notin R$

Hence, R is not symmetric.

$(1, 2) \in R$ and $(2, 3) \in R$

$\Rightarrow (1, 3) \in R$

Hence, R is transitive.

34. The identity element for the binary operation $*$ defined on $Q \sim \{0\}$ as $a * b =$

$\frac{ab}{2}, \forall a, b \in Q \sim \{0\}$ is

- (A) 1
(B) 0
(C) 2
(D) none of these

Ans: (C)

Given that $a * b = \frac{ab}{2}$, $\forall a, b \in \mathbb{Q} \sim \{0\}$

Let e be the identity element for $*$

Therefore, $a * e = \frac{ae}{2}$ ($a * e = e * a = a$)

$$\Rightarrow a = \frac{ae}{2}$$

$$\Rightarrow e = 2$$

35. If the set A contains 5 elements and the set B contains 6 elements, then the number of one-one and onto mappings from A to B is

- (A) 720
(B) 120
(C) 0
(D) none of these

Ans: We know that, if A and B are two non- empty finite set containing m and n elements respectively, then the number of one-one and onto mapping from A to B is

$n!$ If $m = n$

0, if $m \neq n$

Given that $m = 5$ and $n = 6$

Therefore, $m \neq n$

Number of mapping = 0

36. Let $A = \{1, 2, 3, \dots, n\}$ and $B = \{a, b\}$. Then the number of surjections from A into B is

(A) ${}^n P_2$

(B) $2^n - 2$

(C) $2^n - 1$

(D) None of these

Ans: (A)

Given that:

$$A = \{1, 2, 3, \dots, n\} \text{ and } B = \{a, b\}$$

We know that, if A and B are two non- empty finite set containing m and n elements respectively, then the number of surjections from A to B is

$${}^n C_m \times m!$$

(if $n \geq m$),

0, if $m < n$.

Here, $m = 2$.

$${}^n C_m \times 2!$$

$$\Rightarrow \frac{n!}{2!(n-2)!} \times 2!$$

$$\Rightarrow {}^n C_2$$

$$\Rightarrow \frac{n(n-1)(n-2)!}{2 \times 1(n-2)!} \times 2!$$

$$\Rightarrow n^2 - n$$

37. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{1}{x}, \forall x \in \mathbb{R}$. Then f is

(A) one-one

(B) onto

(C) bijective

(D) f is not defined

Ans: (D)

Given that, $f(x) = \frac{1}{x}, \forall x \in \mathbb{R}$.

For $x = 0$, $f(x)$ is not defined.

Hence, $f(x)$ is a not defined function.

38. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x^2 - 5$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = \frac{x}{x^2 + 1}$. Then $g \circ f$ is:

(A) $\frac{3x^2 - 5}{9x^4 - 30x^2 + 26}$

(B) $\frac{3x^2 - 5}{9x^4 - 6x^2 + 26}$

(C) $\frac{3x^2}{x^4 + 2x^2 - 4}$

(D) $\frac{3x^2}{9x^4 + 30x^2 - 2}$

Ans: (A)

Given that $f(x) = 3x^2 - 5$ and $g(x) = \frac{x}{x^2 + 1}$

$$g \circ f = g\{f(x)\} = g(3x^2 - 5)$$

$$= \frac{3x^2 - 5}{(3x^2 - 5)^2 + 1}$$

$$= \frac{3x^2 - 5}{9x^4 - 30x^2 + 25 + 1}$$

$$= \frac{3x^2 - 5}{9x^4 - 30x^2 + 26}$$

40. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the functions defined by $f(x) = x^3 + 5$. Then $f^{-1}(x)$ is:

(A) $(x + 5)^{\frac{1}{3}}$

(B) $(x - 5)^{\frac{1}{3}}$

(C) $(5 - x)^{\frac{1}{3}}$

(D) $5 - x$

Ans: (B)

Given that, $f(x) = x^3 + 5$

Let $y = x^3 + 5$

$$\Rightarrow x^3 = y - 5$$

$$\Rightarrow x = (y - 5)^{\frac{1}{3}}$$

$$\Rightarrow f^{-1}(x) = (x - 5)^{\frac{1}{3}}$$

41. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be the bijective functions. Then $(g \circ f)^{-1}$ is

(A) $f^{-1} \circ g^{-1}$

(B) $f \circ g$

(C) $g^{-1} \circ f^{-1}$

(D) $g \circ f$

Ans: (A)

Given that $f: A \rightarrow B$ and $g: B \rightarrow C$ be the bijective functions.

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

42. Let $f: \mathbb{R} - \left\{ \frac{3}{5} \right\} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{3x+2}{5x-3}$. Then

(A) $f^{-1}(x) = f(x)$

(B) $f^{-1}(x) = -f(x)$

(C) $(f \circ f)x = -x$

(D) $f^{-1}(x) = \frac{1}{19}f(x)$

Ans: (A)

Given that, $f(x) = \frac{3x + 2}{5x - 3}$

Let $y = \frac{3x + 2}{5x - 3}$

$3x + 2 = 5xy - 3y$

$\Rightarrow x(3 - 5y) = -3y - 2$

$x = \frac{3x + 2}{5x - 3}$

$\Rightarrow f^{-1}(x) = \frac{3x + 2}{5x - 3}$

$\Rightarrow f^{-1}(x) = f(x)$

43. Let $f:[0,1] \rightarrow [0,1]$ be defined by $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 1 - x, & \text{if } x \text{ is irrational} \end{cases}$

Then $(f \circ f)x$ is

(A) Constant

(B) $1+x$

(C) x

(D) none of these

Ans: (C)

Given that $f:[0,1] \rightarrow [0,1]$ be defined by

$$f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 1-x, & \text{if } x \text{ is irrational} \end{cases}$$

Therefore,

$$(f \circ f)x = f(f(x)) = x$$

44. Let $f : [2, \infty) \rightarrow \mathbb{R}$ be the function defined by $f(x) = x^2 - 4x + 5$, then the range of f is

(A) $\mathbb{R}$

(B) $[1, \infty)$

(C) $[4, \infty)$

(D) $[5, \infty)$

Ans: (B)

Given that, $f(x) = x^2 - 4x + 5$,

$$y = x^2 - 4x + 5$$

$$\Rightarrow y = x^2 - 4x + 4 + 1 = (x-2)^2 + 1$$

$$\Rightarrow (x-2)^2 = y - 1$$

$$\Rightarrow (x-2) = \sqrt{y-1}$$

$$\Rightarrow x = 2 + \sqrt{y-1}$$

Therefore, $(y-1) \geq 0$, $y \geq 1$.

Range = $[1, \infty)$

45. Let $f : \mathbb{N} \rightarrow \mathbb{R}$ be the function defined by $f(x) = \frac{2x-1}{2}$ and $g : \mathbb{Q} \rightarrow \mathbb{R}$ be another function defined by $g(x) = x + 2$. Then $(g \circ f) \frac{3}{2}$ is

(A) 1

(B) 3

(C) $\frac{7}{2}$

(D) none of these

Ans: (B)

Given that $f(x) = \frac{2x-1}{2}$ and $g(x) = x+2$

$$(g \circ f)^{\frac{3}{2}} = g \left[f \left(\frac{3}{2} \right) \right] = g \left(\frac{2 \times \frac{3}{2} - 1}{2} \right)$$

$$= g(1) = 1 + 2 = 3$$

46. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 2x : x > 3 \\ x^2 : 1 < x \leq 3 \\ 3x : x \leq 1 \end{cases}$$

Then $f(-1) + f(2) + f(4)$ is

(A) 9

(B) 14

(C) 5

(D) none of these

Ans: (A)

Given that,

$$f(x) = \begin{cases} 2x : x > 3 \\ x^2 : 1 < x \leq 3 \\ 3x : x \leq 1 \end{cases}$$

$$f(-1) + f(2) + f(4)$$

$$= 3(-1) + 2^2 + (2 \times 4)$$

$$= -3 + 4 + 8$$

$$= 9$$

47. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = \tan x$. Then $f^{-1}(1)$ is

(A) 4

(B) $\left\{n\pi + \frac{\pi}{4} : n \in \mathbb{Z}\right\}$

(C) does not exist

(D) none of these

Ans: (A)

Given that: $f(x) = \tan x$

Let $y = \tan x$

$$\Rightarrow x = \tan^{-1}y$$

$$\Rightarrow f^{-1}(x) = \tan^{-1}x$$

$$\Rightarrow f^{-1}(1) = \tan^{-1}1$$

$$\Rightarrow \tan^{-1}\tan\frac{\pi}{4} = \frac{\pi}{4} \text{ [Since } \tan\frac{\pi}{4} = 1 \text{]}$$

48. Let the relation R be defined in $\mathbb{N}$ by aRb if $2a + 3b = 30$. Then $R = \underline{\hspace{2cm}}$.

Ans: Given that, $2a + 3b = 30$

$$3b = 30 - 2a$$

$$b = \frac{30 - 2a}{3}$$

For $a = 3$, $b = 8$

$a = 6$, $b = 6$

$a = 9$, $b = 4$

$a = 12$, $b = 2$

$$R = \{(3,8), (6,6), (9,4), (12,2)\}$$

49. Let the relation R be defined on the set $A = \{1, 2, 3, 4, 5\}$ by $R = \{(a, b) : |a^2 - b^2| < 8\}$. Then R is given by _____.

Ans: Given $A = \{1, 2, 3, 4, 5\}$

$$R = \{(a, b) : |a^2 - b^2| < 8\}$$

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 3), (3, 4), (4, 4), (5, 5)\}.$$

50. Let $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(2, 3), (5, 1), (1, 3)\}$. Then $g \circ f =$ _____ and $f \circ g =$ _____.

Ans:

Given that

$$f = \{(1, 2), (3, 5), (4, 1)\} \text{ and } g = \{(2, 3), (5, 1), (1, 3)\}.$$

$$g \circ f(1) = g\{f(1)\} = g(2) = 3$$

$$g \circ f(3) = g\{f(3)\} = g(5) = 1$$

$$g \circ f(4) = g\{f(4)\} = g(1) = 3$$

$$g \circ f = \{(1, 3), (3, 1), (4, 3)\}$$

$$f \circ g(x) = f\{g(x)\}$$

$$f \circ g(2) = f\{g(2)\} = f(3) = 5$$

$$f \circ g(5) = f\{g(5)\} = f(1) = 2$$

$$f \circ g(1) = f\{g(1)\} = f(3) = 5$$

$$f \circ g = \{(2, 5), (5, 2), (1, 5)\}$$

51. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{x}{\sqrt{1+x^2}}$, then $(f \circ f \circ f)(x) =$ _____.

Ans:

$$\text{Given that, } f(x) = \frac{x}{\sqrt{1+x^2}}$$

$$(f \circ f \circ f)(x) = f[f\{f(x)\}]$$

$$\begin{aligned}
 &= f \left[f \left(\frac{x}{\sqrt{1+x^2}} \right) \right] \\
 &= f \left(\frac{\frac{x}{\sqrt{1+x^2}}}{\sqrt{1+\frac{x^2}{1+x^2}}} \right) \\
 &= f \left(\frac{x\sqrt{1+x^2}}{\sqrt{1+x^2}(\sqrt{2x^2+1})} \right) \\
 &= f \left(\frac{x}{\sqrt{1+2x^2}} \right) f \left(\frac{\frac{x}{\sqrt{1+2x^2}}}{\frac{\sqrt{1+x^2+x^2}}{\sqrt{1+x^2}}} \right) \\
 &= f \left(\frac{x}{\sqrt{1+2x^2}} \right) \\
 &= \frac{\frac{x}{\sqrt{1+2x^2}}}{\sqrt{1+\frac{x^2}{1+x^2}}} \\
 &= \frac{x\sqrt{1+2x^2}}{\sqrt{1+2x^2}\sqrt{1+3x^2}} \\
 &= \frac{x}{\sqrt{1+3x^2}}
 \end{aligned}$$

52. If $f(x) = [4 - (x - 7)^3]$, then $f^{-1}(x) = \underline{\hspace{2cm}}$.

Ans: Given that, $f(x) = [4 - (x - 7)^3]$

Let $y = [4 - (x - 7)^3]$

$$(x - 7)^3 = 4 - y$$

$$\Rightarrow (x - 7) = (4 - y)^{1/3}$$

$$\Rightarrow x = 7 + (4 - y)^{1/3}$$

$$\Rightarrow f^{-1}(x) = 7 + (4 - x)^{1/3}$$

True/False:

53. Let $R = \{(3, 1), (1, 3), (3, 3)\}$ be a relation defined on the set $A = \{1, 2, 3\}$. Then R is symmetric, transitive but not reflexive.

Ans: The given statement is false.

Given that, $R = \{(3,1), (1,3), (3, 3)\}$ be defined on the set $A = \{1, 2, 3\}$

$$(1,1) \notin R$$

So, R is not reflexive.

$$(3,1) \notin R, (1,3) \in R$$

Hence, R is symmetric.

$$\text{Since, } (3,1) \in R, (1,3) \in R$$

$$\text{But } (1,1) \notin R$$

Hence, R is not transitive.

54. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = \sin(3x+2) \forall x \in \mathbb{R}$. Then f is invertible.

Ans: The given statement is false.

Given $f(x) = \sin(3x + 2) \forall x \in \mathbb{R}$ is not one-one function for all $x \in \mathbb{R}$.

So, f is not invertible.

55. Every relation which is symmetric and transitive is also reflexive.

Ans: The given statement is false.

Let R be a relation defined by:

$R = \{ (1,2), (2,1), (1,1), (2,2) \}$ on the set $A = \{1,2,3\}$

It is clear that $(3,3) \in R$. So, it is not reflexive.

56. An integer m is said to be related to another integer n if m is an integral multiple of n . This relation in $\mathbb{Z}$ is reflexive, symmetric and transitive.

Ans: The given statement is false.

The given relation is reflexive and transitive but not symmetric.

57. Let $A = \{0, 1\}$ and N be the set of natural numbers. Then the mapping $f : N \rightarrow A$ defined by $f(2n-1) = 0$, $f(2n) = 1$, $\forall n \in N$, is onto.

Ans: The given statement is true.

Given, $A = \{0,1\}$

$f(2n-1) = 0$, $f(2n) = 1, \forall n \in N$

So, the mapping $f: N \rightarrow A$ is onto.

58. The relation R on the set $A = \{1, 2, 3\}$ defined as $R = \{(1, 1), (1, 2), (2, 1), (3, 3)\}$ is reflexive, symmetric and transitive.

Ans: The given statement is false.

Given that $R = \{(1,1), (1,2), (2,1), (3,3)\}$

$(2,2) \notin R$

So, R is not reflexive.

59. The composition of functions is commutative.

Ans: The given statement is false.

Let $f(x) = x^2$

And $g(x) = x + 1$

$f \circ g(x) = f\{g(x)\} = f(x + 1)$

$= (x+1)^2 = x^2 + 2x + 1$

$g \circ f(x) = g\{f(x)\} = g(x^2) = x^2 + 1$

Therefore, $f \circ g(x) \neq g \circ f(x)$.

60. The composition of functions is associative.

Ans: The given statement is true.

Let $f(x) = x$, $g(x) = x + 1$

And $h(x) = 2x - 1$

Then, $f \circ \{g \circ h(x)\}$

$$= f[g\{h(x)\}]$$

$$= f\{g(2x - 1)\}$$

$$= f(2x - 1) + 1$$

$$= f(2x) = 2x$$

$$(f \circ g)h(x) = (f \circ g)\{h(x)\}$$

$$= (f \circ g)(2x - 1)$$

$$= f\{g(2x - 1)\}$$

$$= f(2x - 1 + 1)$$

$$= f(2x) = 2x$$

61. Every function is invertible.

Ans: The given statement is false.

Only bijective functions are invertible.

62. A binary operation on a set has always the identity element.

Ans: The given statement is false.

‘+’ is a binary operation on the set N but it has no identity element.